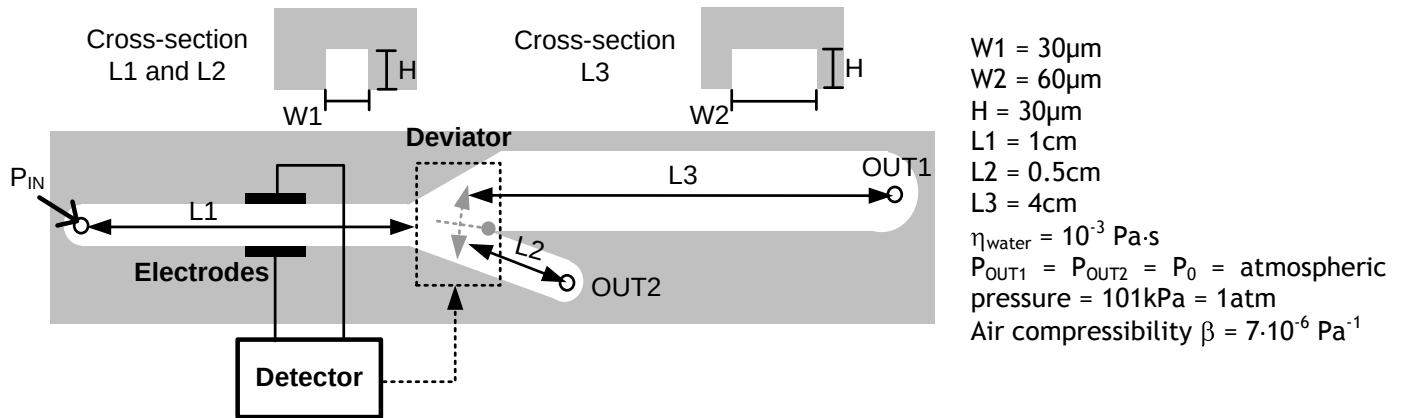


Politecnico di Milano
Biochip A.A. 2013/2014 - M. Carminati
Appello 22.07.2014

*Test duration: 2 hours - No support material (textbooks, laptops, smartphones, notes) admitted.
 All the answers must be motivated. Beyond the final numerical results, all the expressions must be provided.*

Exercise 1

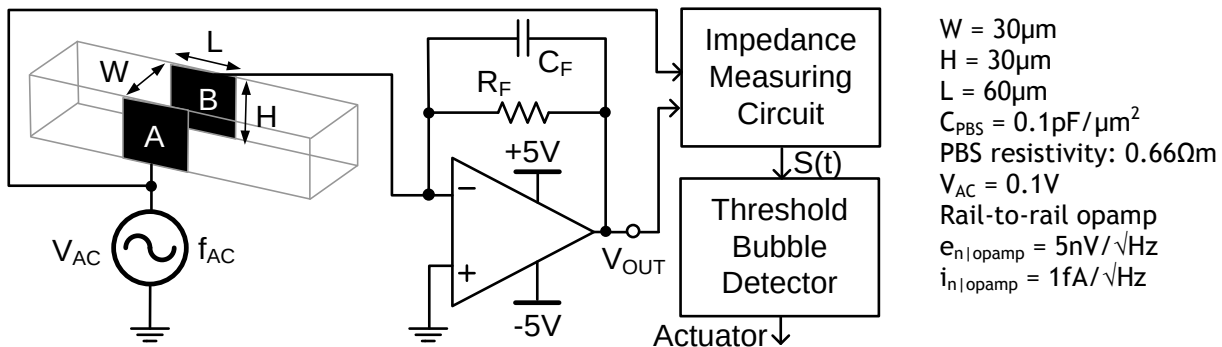
Consider the following microfluidic system for removing air bubbles from the sample: the detector senses the presence of a bubble and drives the actuator to deviate it to OUT2 while the de-bubbled liquid exits from OUT1.



- Determine P_{IN} in order to have a fluid speed of $1 \mu\text{m}/\text{ms}$ in the electrodes section (no deviator).
- Determine P_{IN} as in case (a) but for a blood sample instead of water/PBS.
- Determine the time required to fill the whole channel starting from completely empty condition [P_{IN} as in (a)].
- If the outlet OUT2 is closed and the section L2 completely filled with air, what is the associated time constant?
- Without the deviator, what is the efficiency of the device (i.e. the relative amount of lost sample at OUT2)? Propose a solution to improve the efficiency.
- Propose an implementation of the deviating actuator.

Exercise 2

Now focus on the circuit of the bubble detector of Ex. 1:



- If the channel is filled with PBS (neglecting all parasitic paths and fringing fields) determine the detector equivalent impedance (between electrodes A and B) and draw it both in the Bode Plot (magnitude and phase) and in the Cole-Cole Plot formats.
- Draw the detailed scheme of the lock-in impedance measuring block (choosing which parts are required in this case of bubble detection).
- Determine the value of a suitable sensing frequency f_{AC} , considering also a stray bridge capacitance C_S between A and B of 1pF .
- Considering the minimum C_F , determine the value of R_F and plot $V_{OUT}(t)$ in the absence of bubbles, at f_{AC} of (c).
- Draw the signal $S(t)$ when a bubble of volume equal to twice the sensing volume passes through the sensor at $1 \mu\text{m}/\text{ms}$ speed, quoting both amplitude [V] and time [sec] axis's. Consequently choose the proper value of bandwidth of the lock-in filter and the value of the bubble detection threshold.
- If $R_F = 100\text{k}\Omega$, $C_F = 0.5\text{pF}$, $C_S = 1\text{pF}$ determine and plot the input noise current spectral density (neglect the V_{AC} generator noise and replace C_{DL} with a short circuit).
- If the rms noise of the whole detection system were 20ppm , what would be the minimum volume of a detectable air bubble (suggestion: consider the volume fraction).
- Is it possible to distinguish between an air bubble and a biological cell with the same size? (motivate the answer).